loop makes an angle θ with the magnetic field as shown in Figure 3.57.

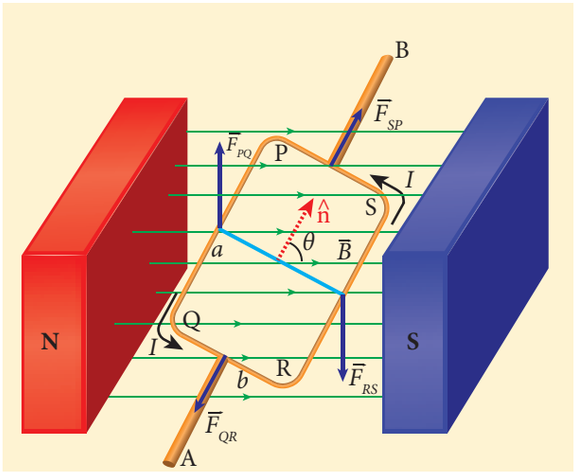


Figure 3.57 Rectangular coil placed in a magnetic field

The magnitude of the magnetic force acting on the current-carrying arm PQ is $F_{PQ} = IaB\sin\left(\pi/2\right) = IaB$. The direction of the force is found using right hand cork screw rule and its direction is upwards.

The magnitude of the force on the arm QR is $F_{QR} = IbB\sin\left(\pi/2 - \theta\right) = IbB\cos\theta$ and its direction is as shown in Figure 3.57.

The magnitude of the force on the arm RS is $F_{RS} = IaB\sin\left(\pi/2\right) = IaB$ and its direction is downwards.

The magnitude of the force acting on the arm SP is $F_{SP} = IbB\sin\left(\pi/2 + \theta\right) = IbB\cos\theta$ and its direction is also as shown in the Figure 3.57.

Since the forces F_{QR} and F_{SP} are equal, opposite and collinear, they cancel each other. But the forces F_{PQ} and F_{RS} , which are equal in magnitude and opposite in direction, are not acting along same straight line. Therefore, F_{PQ} and F_{RS} constitute a couple which exerts a torque on the loop.

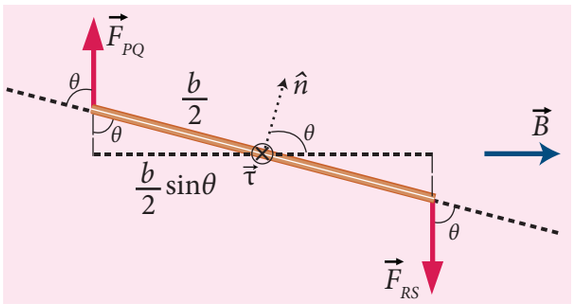


Figure 3.58 Side view of current loop

The magnitude of torque acting on the arm PQ about AB is $\tau_{PQ} = \left(\frac{b}{2}\sin\theta\right)IaB$ and it points in the direction of AB. The magnitude of the torque acting on the arm RS about AB is $\tau_{RS} = \left(\frac{b}{2}\sin\theta\right)IaB$ and points also in the same direction AB as shown in Figure 3.58.

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The total torque acting on the entire loop about an axis AB is given by

$$\tau = \left(\frac{b}{2}\sin\theta\right)F_{PQ} + \left(\frac{b}{2}\sin\theta\right)F_{RS}$$
$$= Ia(b\sin\theta)B$$

$$\tau = IAB\sin\theta \quad \text{along the direction AB}$$

In vector form,

$$\vec{\tau} = \left(I\vec{A}\right)\times\vec{B}$$

The above equation can also be written in terms of magnetic dipole moment

$$\vec{\tau} = \vec{p}_m \times \vec{B} \text{ where } \vec{p}_m = I\vec{A}$$

The tendency of the torque is to rotate the loop so as to align its normal vector with the direction of the magnetic field.

If there are N turns in the rectangular loop, then the torque is given by

$$\tau = NIAB\sin\theta$$

UNIT-3(XII-Physics_Vol-1).indd 180 14-02-2020 20:04:52